

Fall 2025 Math 261 Recitation 1

1. [Section 1.2, 19] Let $V = \{(a_1, a_2) : a_1, a_2 \in \mathbb{R}\}$. Define addition of elements of V coordinatewise, and for (a_1, a_2) in V and $\alpha \in \mathbb{R}$, define

$$\alpha(a_1, a_2) = \begin{cases} (0, 0) & \text{if } \alpha = 0 \\ \left(\alpha a_1, \frac{a_2}{\alpha}\right) & \text{if } \alpha \neq 0. \end{cases}$$

Is V a vector space over $\mathbb{R}$ with these operations? Justify your answer.

Solution: We must have $(\alpha + \beta)x = \alpha x + \beta x$ for all $\alpha, \beta \in \mathbb{R}$ and $x \in V$. Pick $\alpha \neq 0$ and $\beta \neq 0$ with $\alpha + \beta \neq 0$. Then

$$\begin{aligned} (\alpha + \beta)(a_1, a_2) &= \left((\alpha + \beta)a_1, \frac{a_2}{\alpha + \beta}\right) \\ \alpha(a_1, a_2) + \beta(a_1, a_2) &= \left(\alpha a_1, \frac{a_2}{\alpha}\right) + \left(\beta a_1, \frac{a_2}{\beta}\right) = \left((\alpha + \beta)a_1, \frac{a_2}{\alpha} + \frac{a_2}{\beta}\right) \end{aligned}$$

The first coordinates match, but the second coordinates are $\frac{a_2}{\alpha + \beta}$ and $\frac{a_2}{\alpha} + \frac{a_2}{\beta}$, which are not equal in general. So the distributive law fails. Thus V is not a vector space over $\mathbb{R}$.

2. [Fall 2024 MT1, Q1]

Consider the first quadrant $V = \{(x, y) \in \mathbb{R}^2 : x, y > 0\}$ in the plane together with the operations $\oplus : V \times V \rightarrow V$ and $\otimes : \mathbb{R} \times V \rightarrow V$ given by

$$(x_1, y_1) \oplus (x_2, y_2) = (x_1 x_2, y_1 y_2), \quad \alpha \otimes (x, y) = (x^\alpha, y^\alpha)$$

You are given that $(V, \oplus)$ is an abelian group with identity $1_V = (1, 1)$. Show that $(V, \oplus, \otimes)$ is a vector space over $\mathbb{R}$.

Solution: We verify the remaining vector space axioms.

- (a) $1 \otimes (x, y) = (x^1, y^1) = (x, y)$
- (b) $\alpha \otimes (\beta \otimes (x, y)) = \alpha \otimes (x^\beta, y^\beta) = ((x^\beta)^\alpha, (y^\beta)^\alpha) = (x^{\alpha\beta}, y^{\alpha\beta}) = (\alpha\beta) \otimes (x, y)$
- (c) $(\alpha + \beta) \otimes (x, y) = (x^{\alpha+\beta}, y^{\alpha+\beta}) = (x^\alpha x^\beta, y^\alpha y^\beta) = (x^\alpha, y^\alpha) \oplus (x^\beta, y^\beta) = (\alpha \otimes (x, y)) \oplus (\beta \otimes (x, y))$
- (d) $\alpha \otimes ((x_1, y_1) \oplus (x_2, y_2)) = \alpha \otimes (x_1 x_2, y_1 y_2) = ((x_1 x_2)^\alpha, (y_1 y_2)^\alpha) = (x_1^\alpha x_2^\alpha, y_1^\alpha y_2^\alpha) = (x_1^\alpha, y_1^\alpha) \oplus (x_2^\alpha, y_2^\alpha) = (\alpha \otimes (x_1, y_1)) \oplus (\alpha \otimes (x_2, y_2))$

All vector space axioms are satisfied.

3. [Section 1.3, 8(b,c)] Determine whether the following sets are subspaces of $\mathbb{R}^3$ under the operations of addition and scalar multiplication defined on $\mathbb{R}^3$. Justify your answers.

(a) $W_1 = \{(a_1, a_2, a_3) \in \mathbb{R}^3 : a_1 = a_3 + 2\}$

(b) $W_2 = \{(a_1, a_2, a_3) \in \mathbb{R}^3 : 2a_1 - 7a_2 + a_3 = 0\}$

Solution:

(a) W_1 is not a subspace. The zero vector $(0, 0, 0)$ does not satisfy $0 = 0 + 2$, so $0 \notin W_1$.

(b) W_2 is a subspace because we have

- $0 = 2 \cdot 0 - 7 \cdot 0 + 0$, so $(0, 0, 0) \in W_2$
- If $(a_1, a_2, a_3), (b_1, b_2, b_3) \in W_2$, then $2a_1 - 7a_2 + a_3 = 0$ and $2b_1 - 7b_2 + b_3 = 0$. Their sum satisfies $2(a_1 + b_1) - 7(a_2 + b_2) + (a_3 + b_3) = 0$. Thus W_2 is closed under addition.
- If $(a_1, a_2, a_3) \in W_2$ and $c \in \mathbb{R}$, then $2(ca_1) - 7(ca_2) + (ca_3) = c(2a_1 - 7a_2 + a_3) = 0$, so $c(a_1, a_2, a_3) \in W_2$. Hence W_2 is closed under scalar multiplication.

4. [Section 1.3, 10] Prove that $W_1 = \{(a_1, a_2, \dots, a_n) \in \mathbb{F}^n : a_1 + a_2 + \dots + a_n = 0\}$ is a subspace of $\mathbb{F}^n$, but $W_2 = \{(a_1, a_2, \dots, a_n) \in \mathbb{F}^n : a_1 + a_2 + \dots + a_n = 1\}$ is not.

Solution:

For W_1 :

- $0 + 0 + \dots + 0 = 0$, so $0 \in W_1$
- If $\mathbf{a}, \mathbf{b} \in W_1$, then $\sum a_i = 0$ and $\sum b_i = 0$, so $\sum(a_i + b_i) = 0$, thus $\mathbf{a} + \mathbf{b} \in W_1$
- If $\mathbf{a} \in W_1$ and $c \in \mathbb{F}$, then $\sum(ca_i) = c \sum a_i = 0$, so $c\mathbf{a} \in W_1$

For W_2 : The zero vector $0 + \dots + 0 = 0 \neq 1$, so $0 \notin W_2$, thus not a subspace.

5. [Section 1.3, 13] Let S be a nonempty set and $\mathbb{F}$ a field. Prove that for any $s_0 \in S$, the set $\{f \in \mathcal{F}(S, \mathbb{F}) : f(s_0) = 0\}$, is a subspace of $\mathcal{F}(S, \mathbb{F})$.

Solution: Let $W = \{f \in \mathcal{F}(S, \mathbb{F}) : f(s_0) = 0\}$.

- The zero function $0(s) = 0$ for all $s \in S$, so $0(s_0) = 0$, thus $0 \in W$
- If $f, g \in W$, then $f(s_0) = 0$ and $g(s_0) = 0$, so $(f + g)(s_0) = f(s_0) + g(s_0) = 0$, thus $f + g \in W$
- If $f \in W$ and $c \in \mathbb{F}$, then $(cf)(s_0) = c \cdot f(s_0) = 0$, thus $cf \in W$

Therefore, W is a subspace.

6. [Section 1.3, 19] Let W_1 and W_2 be subspaces of a vector space V . Prove that $W_1 \cup W_2$ is a subspace of V if and only if $W_1 \subseteq W_2$ or $W_2 \subseteq W_1$.

Solution:

- ($\Leftarrow$) If $W_1 \subseteq W_2$, then $W_1 \cup W_2 = W_2$, which is a subspace. Similarly if $W_2 \subseteq W_1$.
- ($\Rightarrow$) Suppose $W_1 \cup W_2$ is a subspace but $W_1 \not\subseteq W_2$ and $W_2 \not\subseteq W_1$. Then there exist $x \in W_1 \setminus W_2$ and $y \in W_2 \setminus W_1$. Since $x, y \in W_1 \cup W_2$ and $W_1 \cup W_2$ is a subspace, $x + y \in W_1 \cup W_2$. So $x + y \in W_1$ or $x + y \in W_2$. If $x + y \in W_1$, then $y = (x + y) - x \in W_1$ (since W_1 is a subspace), contradiction. If $x + y \in W_2$, then $x = (x + y) - y \in W_2$ (since W_2 is a subspace), contradiction. Therefore, we must have $W_1 \subseteq W_2$ or $W_2 \subseteq W_1$.

7. [Section 1.3, 26] In $M_{m \times n}(\mathbb{F})$ define

$$W_1 = \{A \in M_{m \times n}(\mathbb{F}) : A_{ij} = 0 \text{ whenever } i > j\}$$

$$W_2 = \{A \in M_{m \times n}(\mathbb{F}) : A_{ij} = 0 \text{ whenever } i \leq j\}$$

Show that $M_{m \times n}(\mathbb{F}) = W_1 \oplus W_2$.

Solution: First, $W_1 \cap W_2 = \{0\}$ because if $A \in W_1 \cap W_2$, then for all i, j : if $i > j$, $A_{ij} = 0$ (from W_1), and if $i \leq j$, $A_{ij} = 0$ (from W_2), so all entries are 0. For any $A \in M_{m \times n}(\mathbb{F})$, define $B \in W_1$ and $C \in W_2$ by:

$$B_{ij} = \begin{cases} A_{ij} & \text{if } i \leq j \\ 0 & \text{if } i > j \end{cases}, \quad C_{ij} = \begin{cases} 0 & \text{if } i \leq j \\ A_{ij} & \text{if } i > j \end{cases}$$

Then $A = B + C$ with $B \in W_1$, $C \in W_2$. Therefore, $M_{m \times n}(\mathbb{F}) = W_1 \oplus W_2$.

8. [Section 1.3, 28] A matrix M is called skew-symmetric if $M^t = -M$. Let $\mathbb{F}$ be a field.

- (a) Prove that the set W_1 of all skew-symmetric $n \times n$ matrices with entries from $\mathbb{F}$ is a subspace of $M_{n \times n}(\mathbb{F})$.
- (b) Now assume that $\mathbb{F}$ is not of characteristic 2, and let W_2 be the subspace of $M_{n \times n}(\mathbb{F})$ consisting of all symmetric $n \times n$ matrices. Prove that $M_{n \times n}(\mathbb{F}) = W_1 \oplus W_2$.

Solution:

- (a) Let $W_1 = \{A \in M_{n \times n}(\mathbb{F}) : A^t = -A\}$.

- Clearly $0 \in W_1$.
- If $A, B \in W_1$, then $A^t = -A$ and $B^t = -B$. Then

$$(A + B)^t = A^t + B^t = -A - B = -(A + B)$$

So $A + B \in W_1$.

- If $A \in W_1$ and $c \in \mathbb{F}$, then

$$(cA)^t = cA^t = c(-A) = -cA$$

So $cA \in W_1$.

Therefore, W_1 is a subspace of $M_{n \times n}(\mathbb{F})$.

- (b) Let $W_2 = \{A \in M_{n \times n}(\mathbb{F}) : A^t = A\}$ be the subspace of symmetric matrices. If $A \in W_1 \cap W_2$, then $A^t = -A$ and $A^t = A$, so $A = -A$. Therefore $2A = 0$. Since $\text{char}(\mathbb{F}) \neq 2$, we have $A = 0$. Therefore $W_1 \cap W_2 = \{0\}$: Now we will show $M_{n \times n}(\mathbb{F}) = W_1 + W_2$. For any $A \in M_{n \times n}(\mathbb{F})$, define:

$$B = \frac{1}{2}(A + A^t) \quad \text{and} \quad C = \frac{1}{2}(A - A^t)$$

Then:

- $B^t = \frac{1}{2}(A + A^t)^t = \frac{1}{2}(A^t + A) = B$, so $B \in W_2$
- $C^t = \frac{1}{2}(A - A^t)^t = \frac{1}{2}(A^t - A) = -C$, so $C \in W_1$
- $A = B + C$

Therefore, $M_{n \times n}(\mathbb{F}) = W_1 \oplus W_2$.

Fall 2025 Math 261 Recitation 2

Problem 1 (1.4-12) Show that a subset W of a vector space V is a subspace of V if and only if $\text{span}(W) = W$.

Solution:

($\Rightarrow$) If W is a subspace, then by definition W is closed under linear combinations, so $\text{span}(W) \subseteq W$.

Also $W \subseteq \text{span}(W)$ since every vector is a trivial linear combination of itself. Thus $\text{span}(W) = W$.

($\Leftarrow$) If $\text{span}(W) = W$ then by Theorem 1.5 $\text{span}(W)$ is a subspace, so W is a subspace.

Problem 2 (1.4-13) Show that if S_1 and S_2 are subsets of a vector space V such that $S_1 \subseteq S_2$, then $\text{span}(S_1) \subseteq \text{span}(S_2)$. In particular, if $S_1 \subseteq S_2$ and $\text{span}(S_1) = V$, deduce that $\text{span}(S_2) = V$.

Solution: Let $v \in \text{span}(S_1)$. Then $v = a_1 s_1 + \cdots + a_n s_n$ for some $s_i \in S_1$ and scalars a_i . Since $S_1 \subseteq S_2$, we have $s_i \in S_2$, so $v \in \text{span}(S_2)$. Thus $\text{span}(S_1) \subseteq \text{span}(S_2)$. If $\text{span}(S_1) = V$ and $S_1 \subseteq S_2$, then:

$$V = \text{span}(S_1) \subseteq \text{span}(S_2) \subseteq V$$

So $\text{span}(S_2) = V$.

Problem 3 (1.5-1(a,d,e)) Label the following statements as true or false.

1. If S is a linearly dependent set, then each vector in S is a linear combination of other vectors in S .
2. Subsets of linearly dependent sets are linearly dependent.
3. Subsets of linearly independent sets are linearly independent.

Solution:

1. **False.** Counterexample: $S = \{(1, 0), (2, 0), (0, 1)\}$ in $\mathbb{R}^2$. The set is linearly dependent since $(1, 0)$ and $(2, 0)$ are dependent, but $(0, 1)$ is not a linear combination of the other two.
2. **False.** Counterexample: $S = \{(1, 0), (2, 0), (0, 1)\}$ is linearly dependent, but the subset $\{(0, 1)\}$ is linearly independent.
3. **True.** If a subset of a linearly independent set were linearly dependent, then the original set would also be linearly dependent by Theorem 1.6.

Problem 4 (1.5-13 (a)) Let V be a vector space over a field of characteristic not equal to two. Let u and v be distinct vectors in V . Prove that $\{u, v\}$ is linearly independent if and only if $\{u+v, u-v\}$ is linearly independent.

Solution: ($\Rightarrow$) Suppose $\{u, v\}$ is linearly independent. Assume $a(u+v) + b(u-v) = 0$. Then $(a+b)u + (a-b)v = 0$. Since $\{u, v\}$ is linearly independent, $a+b=0$ and $a-b=0$. Adding gives $2a=0$, so $a=0$ (since $\text{char} \neq 2$), and then $b=0$. Thus $\{u+v, u-v\}$ is linearly independent.

($\Leftarrow$) Suppose $\{u+v, u-v\}$ is linearly independent. Assume $au+bv=0$. Write $u = \frac{1}{2}[(u+v) + (u-v)]$ and $v = \frac{1}{2}[(u+v) - (u-v)]$. Then $au+bv = \frac{a+b}{2}(u+v) + \frac{a-b}{2}(u-v) = 0$. Since $\{u+v, u-v\}$ is linearly independent, $\frac{a+b}{2}=0$ and $\frac{a-b}{2}=0$, so $a=b=0$.

Problem 5 (1.5-14) Prove that a set S is linearly dependent if and only if $S = \{0\}$ or there exist distinct vectors $v, u_1, u_2, \dots, u_n$ in S such that v is a linear combination of $u_1, u_2, \dots, u_n$.

Solution: ($\Rightarrow$) If S is linearly dependent, then there exist distinct vectors $v_1, \dots, v_k \in S$ and scalars $c_1, \dots, c_k$ not all zero such that $c_1v_1 + \dots + c_kv_k = 0$. If $S \neq \{0\}$, then at least one $c_i \neq 0$. Without loss of generality, assume $c_1 \neq 0$. Then $v_1 = -\frac{c_2}{c_1}v_2 - \dots - \frac{c_k}{c_1}v_k$, so v_1 is a linear combination of the other vectors. ($\Leftarrow$) If $S = \{0\}$, then $1 \cdot 0 = 0$ is a nontrivial representation. Otherwise, if $v = a_1u_1 + \dots + a_nu_n$ for distinct vectors in S , then $(-1)v + a_1u_1 + \dots + a_nu_n = 0$ is a nontrivial linear combination giving zero.

Problem 6 (1.6-14) Let

$$W_1 = \{(a_1, a_2, a_3, a_4, a_5) \in F^5 : a_1 - a_3 - a_4 = 0\}$$

and

$$W_2 = \{(a_1, a_2, a_3, a_4, a_5) \in F^5 : a_2 = a_3 = a_4 \text{ and } a_1 + a_5 = 0\}.$$

Find bases for W_1 and W_2 , and determine the dimensions of W_1 and W_2 .

Solution:

For W_1 :

The condition is $a_1 = a_3 + a_4$. A general vector in W_1 is:

$$(a_3 + a_4, a_2, a_3, a_4, a_5) = a_2(0, 1, 0, 0, 0) + a_3(1, 0, 1, 0, 0) + a_4(1, 0, 0, 1, 0) + a_5(0, 0, 0, 0, 1).$$

So a basis for W_1 is: $\{(0, 1, 0, 0, 0), (1, 0, 1, 0, 0), (1, 0, 0, 1, 0), (0, 0, 0, 0, 1)\}$. Thus, $\dim(W_1) = 4$.

For W_2 :

The conditions are $a_2 = a_3 = a_4$ and $a_1 + a_5 = 0 \Rightarrow a_5 = -a_1$.

A general vector in W_2 is:

$$(a_1, a_2, a_2, a_2, -a_1) = a_1(1, 0, 0, 0, -1) + a_2(0, 1, 1, 1, 0).$$

So a basis for W_2 is: $\{(1, 0, 0, 0, -1), (0, 1, 1, 1, 0)\}$. Thus, $\dim(W_2) = 2$.

Problem 7 (1.6-30) Let

$$V = M_{2 \times 2}(F), \quad W_1 = \left\{ \begin{bmatrix} a & b \\ c & a \end{bmatrix} \in V : a, b, c \in F \right\},$$

and

$$W_2 = \left\{ \begin{bmatrix} 0 & a \\ -a & b \end{bmatrix} \in V : a, b \in F \right\}.$$

Find the dimensions of the subspaces W_1 , W_2 , $W_1 + W_2$, and $W_1 \cap W_2$.

Solution:

Dimension of W_1 :

A general matrix in W_1 is:

$$\begin{bmatrix} a & b \\ c & a \end{bmatrix} = a \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} + b \begin{bmatrix} 0 & 1 \\ 0 & 0 \end{bmatrix} + c \begin{bmatrix} 0 & 0 \\ 1 & 0 \end{bmatrix}.$$

So a basis for W_1 is:

$$\left\{ \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}, \begin{bmatrix} 0 & 1 \\ 0 & 0 \end{bmatrix}, \begin{bmatrix} 0 & 0 \\ 1 & 0 \end{bmatrix} \right\} = \{u_1, u_2, u_3\}.$$

Thus, $\dim(W_1) = 3$.

Dimension of W_2 :

A general matrix in W_2 is:

$$\begin{bmatrix} 0 & a \\ -a & b \end{bmatrix} = a \begin{bmatrix} 0 & 1 \\ -1 & 0 \end{bmatrix} + b \begin{bmatrix} 0 & 0 \\ 0 & 1 \end{bmatrix}.$$

So a basis for W_2 is:

$$\left\{ \begin{bmatrix} 0 & 1 \\ -1 & 0 \end{bmatrix}, \begin{bmatrix} 0 & 0 \\ 0 & 1 \end{bmatrix} \right\} = \{v_1, v_2\}.$$

Thus, $\dim(W_2) = 2$.

Dimension of $W_1 + W_2$:

Observe that $\mathcal{B}^* = \{v_1\}$ is a basis for $W_1 \cap W_2$ hence $\dim(W_1 \cap W_2) = 1$. Now by using the dimension formula:

$$\dim(W_1 + W_2) = \dim(W_1) + \dim(W_2) - \dim(W_1 \cap W_2),$$

we have $\dim(W_1 + W_2) = 4$. We can also find this dimension by verifying that $\mathcal{B} = \{u_1, u_2, u_3, v_2\}$ is a basis for $W_1 + W_2$. (Observe that $v_1 = u_2 - u_3$).

Problem 8 (Fall 2024 MT2-1(b)) Consider the first quadrant $V = \{(x, y) \in \mathbb{R}^2 : x, y > 0\}$ in the plane together with the operations $\oplus : V \times V \rightarrow V$ and $\otimes : \mathbb{R} \times V \rightarrow V$ given by

$$(x_1, y_1) \oplus (x_2, y_2) = (x_1 x_2, y_1 y_2) \quad \text{and} \quad \alpha \otimes (x, y) = (x^\alpha, y^\alpha)$$

Recall that $(V, \oplus, \otimes)$ is a vector space over $\mathbb{R}$ whose zero vector is the vector $(1, 1)$. Show that $\mathcal{B} = \{(1, 2), (3, 4)\}$ is a basis for V .

Solution:

We first argue that $\mathcal{B}$ is linearly independent. Let $\alpha, \beta \in \mathbb{R}$. Suppose that

$$(\alpha \otimes (1, 2)) \oplus (\beta \otimes (3, 4)) = (1, 1).$$

Then, by the definition of the operations, we have

$$(1^\alpha, 2^\alpha) \oplus (3^\beta, 4^\beta) = (1^\alpha \cdot 3^\beta, 2^\alpha \cdot 4^\beta) = (3^\beta, 2^\alpha 4^\beta) = (1, 1).$$

Since $3^\beta = 1$, we obtain $\beta = 0$. But then we get $2^\alpha = 1$, which implies $\alpha = 0$ as well. Hence $\mathcal{B}$ is linearly independent.

We now argue that $\text{span}(\mathcal{B}) = V$. Clearly $\text{span}(\mathcal{B}) \subseteq V$. Now let $(x, y) \in V$. Consider

$$\alpha = \log_2(y) - 2 \log_3(x) \quad \text{and} \quad \beta = \log_3(x).$$

Then

$$(\alpha \otimes (1, 2)) \oplus (\beta \otimes (3, 4)) = (1^\alpha, 2^\alpha) \oplus (3^\beta, 4^\beta) = (3^\beta, 2^\alpha 4^\beta) = (3^\beta, 2^{\alpha+2\beta}).$$

Substituting the expressions for α and β :

$$(3^{\log_3(x)}, 2^{\log_2(y) - 2 \log_3(x) + 2 \log_3(x)}) = (x, 2^{\log_2(y)}) = (x, y).$$

Hence $(x, y) \in \text{span}(\mathcal{B})$. This shows that $\text{span}(\mathcal{B}) \supseteq V$ and therefore $\text{span}(\mathcal{B}) = V$.

Fall 2025 Math 261 Recitation 3

[2.1-14] 1. Let V and W be vector spaces and $T : V \rightarrow W$ be linear.

- (a) Suppose that T is one-to-one and that S is a subset of V . Prove that S is linearly independent if and only if $T(S)$ is linearly independent.

Solution: ($\Rightarrow$) Let $S = \{v_1, \dots, v_k\}$ be a linearly independent subset of V . Then $T(S) = \{T(v_1), \dots, T(v_k)\}$. Let $\sum c_i T(v_i) = 0$ for some c_i . Then $T(\sum c_i v_i) = 0$. So $\sum c_i v_i \in \text{Ker}(T)$. But since T is one to one $\text{Ker}(T) = \{0\}$. So $\sum c_i v_i = 0$. But we know $\{v_i\}$ is linearly independent. So $c_i = 0$. Hence $T(S)$ is linearly independent.

($\Leftarrow$) Suppose $T(S)$ is linearly independent. Let $\{v_1, \dots, v_k\} \subseteq S$ and suppose $\sum c_i v_i = 0$. Then $T(\sum c_i v_i) = \sum c_i T(v_i) = 0$, so all $c_i = 0$ since $T(S)$ is linearly independent. Thus S is linearly independent.

- (b) Suppose $\beta = \{v_1, v_2, \dots, v_n\}$ is a basis for V and T is one-to-one and onto. Prove that $T(\beta) = \{T(v_1), T(v_2), \dots, T(v_n)\}$ is a basis for W . **Solution:** Since T is one-to-one, $T(\beta)$ is linearly independent by part (a). Since T is onto, for any $w \in W$, there exists $v \in V$ with $T(v) = w$. Write $v = \sum c_i v_i$, then $w = T(v) = \sum c_i T(v_i)$, so $T(\beta)$ spans W . Thus $T(\beta)$ is a basis for W .

[2.1-21] 2. Let $\mathbb{F}$ be any field. Let V consist of all sequences $\{a_n\}$ in $\mathbb{F}$ that have only a finite number of nonzero terms a_n . If $\{a_n\}$ and $\{b_n\}$ are in V and $t \in F$, define

$$\{a_n\} + \{b_n\} = \{a_n + b_n\} \text{ and } t\{a_n\} = \{ta_n\}.$$

Define the functions $T, U : V \rightarrow V$ by

$$T(a_1, a_2, \dots) = (a_2, a_3, \dots) \quad \text{and} \quad U(a_1, a_2, \dots) = (0, a_1, a_2, \dots).$$

T and U are called the *left shift* and *right shift* operators on V , respectively.

- (a) Prove that T and U are linear.

Solution: For T : Let $x = (a_1, a_2, \dots), y = (b_1, b_2, \dots) \in V$ and $c \in F$. Then

$$\begin{aligned} T(cx + y) &= T(ca_1 + b_1, ca_2 + b_2, \dots) \\ &= (ca_2 + b_2, ca_3 + b_3, \dots) \\ &= c(a_2, a_3, \dots) + (b_2, b_3, \dots) = cT(x) + T(y). \text{ Similarly} \\ U(cx + y) &= U(ca_1 + b_1, ca_2 + b_2, \dots) \\ &= (0, ca_1 + b_1, ca_2 + b_2, \dots) \\ &= c(0, a_1, a_2, \dots) + (0, b_1, b_2, \dots) = cU(x) + U(y). \end{aligned}$$

- (b) Prove that T is onto, but not one-to-one.

Solution: T is onto: For any $(b_1, b_2, \dots) \in V$, take $x = (0, b_1, b_2, \dots)$, then $T(x) = (b_1, b_2, \dots)$. T is not one-to-one: $T(1, 0, 0, \dots) = (0, 0, \dots) = T(0, 0, \dots)$, but $(1, 0, 0, \dots) \neq (0, 0, \dots)$.

- (c) Prove that U is one-to-one, but not onto.

Solution: U is one-to-one: If $U(x) = 0$, then $(0, a_1, a_2, \dots) = (0, 0, \dots)$, so all $a_i = 0$, thus $x = 0$. U is not onto: No sequence starting with a nonzero first term is in $\text{Ran}(U)$, since every image under U has 0 as its first term.

[Fall 2018 MT2-Q1]3. Let V be a finite dimensional vector space and let $T : V \rightarrow V$ be a linear transformation.

Prove or disprove the following statements:

(a) If $T^2 = 0$ then $\text{Ran}(T) \subseteq \text{Ker}(T)$.

Solution: True. Pick $v \in \text{Ran}(T)$. Then there exists $u \in V$ such that $T(u) = v$. Since $T^2 = 0$, we have $T^2(u) = T(T(u)) = T(v) = 0$. Thus $v \in \text{Ker}(T)$.

(b) If $T^2 = I_V$ and $v \in V$ then there exists $w \in V$ such that $T(w) = v$.

Solution: True. The map T is invertible because $T \circ T = I_V$. Note that T^{-1} is the map T itself. Therefore, T is onto. This means that for each $v \in V$, there exists $w \in V$ such that $T(w) = v$. Alternatively, given $v \in V$, set $w = T(v)$. Then $v = T^2(v) = T(T(v)) = T(w)$.

(c) If $T^2 = T$ and $T \neq I_V$ then $\text{Ker}(T) \neq \{0\}$.

Solution: True. If $T \neq I_V$ then there exists $v \in V$ such that $T(v) \neq v$. Consider the **nonzero** vector $w = T(v) - v$. We have $T(w) = T(T(v) - v) = T^2(v) - T(v)$. Using $T^2 = T$, we find that $T(w) = 0$. Thus $w \in \text{Ker}(T)$. We conclude that $\text{Ker}(T) \neq \{0\}$.

Definition: Let V be a vector space, and let $T : V \rightarrow V$ be linear. A subspace W of V is said to be T -invariant if $T(x) \in W$ for every $x \in W$.

[2.1-31] 4. Suppose that $V = \text{Ran}(T) \oplus W$ and W is T -invariant.

(a) Prove that $W \subseteq \text{Ker}(T)$.

Solution: Let $w \in W$. Since W is T -invariant, $T(w) \in W$. But also $T(w) \in \text{Ran}(T)$. Since $V = \text{Ran}(T) \oplus W$, we have $\text{Ran}(T) \cap W = \{0\}$, so $T(w) = 0$. Thus $w \in \text{Ker}(T)$.

(b) Show that if V is finite-dimensional, then $W = \text{Ker}(T)$.

Solution: First, by the dimension theorem, $\dim V = \dim \text{Ran}(T) + \dim \text{Ker}(T)$. On the other hand, since $V = \text{Ran}(T) \oplus W$, $\dim V = \dim \text{Ran}(T) + \dim W$. Thus $\dim \text{Ker}(T) = \dim W$. We also know by part (a) that $W \subseteq \text{Ker}(T)$. So $W = \text{Ker}(T)$.

(c) Show by example that the conclusion of (b) is not necessarily true if V is not finite-dimensional.

Solution: Let V be the space of all sequences with finitely many nonzero entries and T be the left shift operator on V . Take $W = \{(0, 0, 0, \dots)\}$. Obviously W is T invariant and $V = \text{Ran}(T) \oplus W$ since T is onto. However $\text{Ker}(T) = \{(a_1, 0, 0, \dots)\} \neq W$.

[2.1-40] 5. Let V be a vector space and W be a subspace of V . Define $T : V \rightarrow V/W$ by $T(v) = v + W$ for $v \in V$.

(a) Prove that T is a linear transformation from V onto V/W and that $\text{Ker}(T) = W$.

Solution: Since $T(v_1 + v_2) = (v_1 + v_2) + W = (v_1 + W) + (v_2 + W) = T(v_1) + T(v_2)$, and $T(cv) = cv + W = c(v + W) = cT(v)$ the map T is linear. Since for any $v + W \in V/W$, $T(v) = v + W$ it is onto. $\text{Ker}(T) = \{v \in V : v + W = W\} = \{v \in V : v \in W\} = W$.

(b) Suppose that V is finite-dimensional. Use (a) and the dimension theorem to derive a formula relating $\dim(V)$, $\dim(W)$, and $\dim(V/W)$.

Solution: By the dimension theorem: $\dim \text{Ker}(T) + \dim \text{Ran}(T) = \dim V$. But $\text{Ker}(T) = W$ and $\text{Ran}(T) = V/W$ (since T is onto), so: $\dim W + \dim(V/W) = \dim V$.

[Fall 2024-MT2-Q1] 6. Consider the first quadrant $V = \{(x, y) \in \mathbb{R}^2 : x, y > 0\}$ in the plane together with the operations $\oplus : V \times V \rightarrow V$ and $\otimes : \mathbb{R} \times V \rightarrow V$ given by

$$(x_1, y_1) \oplus (x_2, y_2) = (x_1 x_2, y_1 y_2) \quad \text{and} \quad \alpha \otimes (x, y) = (x^\alpha, y^\alpha)$$

Recall that $(V, \oplus, \otimes)$ is a vector space over $\mathbb{R}$ whose zero vector is the vector $(1, 1)$. Moreover, recall that $W = \{(x, y) \in V : xy = 1\}$ is a subspace of V . Find an equation that defines the coset $(1, 2) \oplus W$ and then sketch it in the x - y plane.

Solution:

$$\begin{aligned} (1, 2) \oplus W &= \{(1, 2) \oplus (a, b) : (a, b) \in W\} \\ &= \{(a, 2b) \in V : (a, b) \in W\} \\ &= \{(a, 2b) \in V : ab = 1\} \\ &= \{(x, y) \in V : xy = 2\} \end{aligned}$$

and hence the given coset is the hyperbola defined by the equation $xy = 2$ in the first quadrant.

7. Let V be the vector space in the previous question. Define $T : V \rightarrow \mathbb{R}^2$ by $T(x, y) = (\ln x, \ln y)$.

(a) Prove that T is linear.

Solution: For $(x_1, y_1), (x_2, y_2) \in V$ and $c \in \mathbb{R}$:

$$\begin{aligned} T((x_1, y_1) \oplus (x_2, y_2)) &= T(x_1 x_2, y_1 y_2) = (\ln(x_1 x_2), \ln(y_1 y_2)) \\ &= (\ln x_1 + \ln x_2, \ln y_1 + \ln y_2) = T(x_1, y_1) + T(x_2, y_2) \\ T(\alpha \otimes (x, y)) &= T(x^\alpha, y^\alpha) = (\ln(x^\alpha), \ln(y^\alpha)) = (\alpha \ln x, \alpha \ln y) = \alpha T(x, y) \end{aligned}$$

(b) Find $\text{Ker}(T)$ and $\text{Ran}(T)$.

Solution: $\text{Ker}(T) = \{(x, y) \in V : T(x, y) = (0, 0)\} = \{(x, y) : \ln x = 0, \ln y = 0\} = \{(1, 1)\}$

Observe also that for any $(x, y) \in \mathbb{R}^2$, $T(e^x, e^y) = (x, y)$. So $\text{Ran}(T) = \mathbb{R}^2$.

(c) Find $\dim(V)$ by using the dimension theorem.

Solution: By the dimension theorem: $\dim V = \dim(\text{Ker}(T)) + \dim(\text{Ran}(T)) = 0 + 2 = 2$.

[2.2-4] 8. Define $T : M_{2 \times 2}(\mathbb{R}) \rightarrow P_2(\mathbb{R})$ by

$$T \begin{bmatrix} a & b \\ c & d \end{bmatrix} = (a + b) + (2d)x + bx^2.$$

Let

$$\beta = \left\{ \begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix}, \begin{bmatrix} 0 & 1 \\ 0 & 0 \end{bmatrix}, \begin{bmatrix} 0 & 0 \\ 1 & 0 \end{bmatrix}, \begin{bmatrix} 0 & 0 \\ 0 & 1 \end{bmatrix} \right\} = \{v_1, v_2, v_3, v_4\} \quad \text{and} \quad \gamma = \{1, x, x^2\}.$$

Compute $[T]_\beta^\gamma$.

Solution: We compute T applied to each basis vector in β and express the result in terms of the basis γ .

$$1. \quad T(v_1) = 1 + 0 \cdot x + 0 \cdot x^2$$

$$2. \quad T(v_2) = 1 + 0 \cdot x + 1 \cdot x^2$$

$$3. \quad T(v_3) = 0 + 0 \cdot x + 0 \cdot x^2$$

$$4. \quad T(v_4) = 0 + 2 \cdot x + 0 \cdot x^2$$

$$\implies [T]_\beta^\gamma = \begin{pmatrix} 1 & 1 & 0 & 0 \\ 0 & 0 & 0 & 2 \\ 0 & 1 & 0 & 0 \end{pmatrix}$$

9. Recall that if $A = (a_{ij})$ and $B = (b_{jk})$ are matrices with compatible dimensions, then the (i, k) -entry of AB is given by:

$$(AB)_{ik} = \sum_j a_{ij}b_{jk}.$$

Using this, prove that for any real $n \times n$ matrix A ,

$$\text{tr}(AA^t) \geq 0,$$

with equality if and only if $A = 0$.

Solution:

Let $A = (a_{ij}) \in \mathbb{R}^{n \times n}$, so $A^t = (a_{ji})$. Using the matrix multiplication formula, the (i, k) -entry of AA^t is:

$$(AA^t)_{ik} = \sum_{j=1}^n a_{ij}(A^t)_{jk} = \sum_{j=1}^n a_{ij}a_{kj}.$$

In particular, for the diagonal entries ($i = k$):

$$(AA^t)_{ii} = \sum_{j=1}^n a_{ij}a_{ij} = \sum_{j=1}^n a_{ij}^2.$$

Therefore,

$$\text{tr}(AA^t) = \sum_{i=1}^n (AA^t)_{ii} = \sum_{i=1}^n \sum_{j=1}^n a_{ij}^2.$$

Since $a_{ij}^2 \geq 0$ for all i, j , we have

$$\text{tr}(AA^t) \geq 0.$$

Moreover, $\text{tr}(AA^t) = 0$ if and only if $a_{ij}^2 = 0$ for all i, j , i.e., $A = 0$.

MATH 261 Recitation 4

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Section 2.3: Composition of Linear Transformations and Matrix Multiplication

1. (Section 2.3, Exercise 4b) Let $T : P_2(\mathbb{R}) \rightarrow \mathcal{M}_{2 \times 2}(\mathbb{R})$ be the linear transformation defined by

$$T(f(x)) = \begin{pmatrix} f'(0) & 2f(1) \\ 0 & f''(3) \end{pmatrix}.$$

Let β be the standard ordered basis for $P_2(\mathbb{R})$ and α be the standard ordered basis for $\mathcal{M}_{2 \times 2}(\mathbb{R})$. Use Theorem 2.14 to compute the vector $[T(f(x))]_{\alpha}$, where $f(x) = 4 - 6x + 3x^2$.

Solution: From **Theorem 2.14**, we know that for a linear transformation $T : V \rightarrow W$ with ordered bases β and α respectively, the relationship between coordinate vectors is given by:

$$[T(u)]_{\alpha} = [T]_{\beta}^{\alpha} [u]_{\beta}.$$

Step 1: Find the coordinate vector $[f(x)]_{\beta}$. The basis for $P_2(\mathbb{R})$ is $\beta = \{1, x, x^2\}$. Thus

$$[f(x)]_{\beta} = \begin{bmatrix} 4 \\ -6 \\ 3 \end{bmatrix}.$$

Step 2: Find the matrix representation $[T]_{\beta}^{\alpha}$. We evaluate T on each basis vector of β :

$$T(1) = \begin{bmatrix} 0 & 2 \\ 0 & 0 \end{bmatrix} \implies [T(1)]_{\alpha} = \begin{bmatrix} 0 \\ 2 \\ 0 \\ 0 \end{bmatrix}.$$

$$T(x) = \begin{bmatrix} 1 & 2 \\ 0 & 0 \end{bmatrix} \implies [T(x)]_{\alpha} = \begin{bmatrix} 1 \\ 2 \\ 0 \\ 0 \end{bmatrix}.$$

$$T(x^2) = \begin{bmatrix} 0 & 2 \\ 0 & 2 \end{bmatrix} \implies [T(x^2)]_{\alpha} = \begin{bmatrix} 0 \\ 2 \\ 0 \\ 2 \end{bmatrix}.$$

Therefore, the matrix representation is:

$$[T]_{\beta}^{\alpha} = \begin{bmatrix} 0 & 1 & 0 \\ 2 & 2 & 2 \\ 0 & 0 & 0 \\ 0 & 0 & 2 \end{bmatrix} \implies [T(f(x))]_{\alpha} = [T]_{\beta}^{\alpha} [f(x)]_{\beta} = \begin{bmatrix} 0 & 1 & 0 \\ 2 & 2 & 2 \\ 0 & 0 & 0 \\ 0 & 0 & 2 \end{bmatrix} \begin{bmatrix} 4 \\ -6 \\ 3 \end{bmatrix} = \begin{bmatrix} -6 \\ 2 \\ 0 \\ 6 \end{bmatrix}.$$

Recall: Dimension Theorem (Theorem 2.3) and Direct Sums

- **Dimension Theorem:** Let V and W be vector spaces, and let $T : V \rightarrow W$ be linear. If V is finite-dimensional, then $\dim(V) = \text{rank}(T) + \text{nullity}(T)$.
- **Direct Sum:** If W_1 and W_2 are subspaces of V such that $W_1 \cap W_2 = \{0\}$ and $\dim(W_1) + \dim(W_2) = \dim(V)$, then $V = W_1 \oplus W_2$.

2. (Section 2.3, Exercise 16a) Let V be a finite-dimensional vector space, and let $T : V \rightarrow V$ be linear.

- (a) If $\text{rank}(T) = \text{rank}(T^2)$, prove that $\text{Ran}(T) \cap \text{Ker}(T) = \{0\}$. Deduce that $V = \text{Ran}(T) \oplus \text{Ker}(T)$.

Solution:

- (a) Let $y \in \text{Ran}(T) \cap \text{Ker}(T)$. Since $y \in \text{Ran}(T)$, there exists $x \in V$ such that $T(x) = y$. Since $y \in \text{Ker}(T)$, $T(y) = 0$. Substituting $y = T(x)$ into $T(y) = 0$, we get $T(T(x)) = 0 \implies x \in \text{Ker}(T^2)$. From the **Dimension Theorem**, we know that:

$$\dim(V) = \text{rank}(T) + \text{nullity}(T) \quad \text{and} \quad \dim(V) = \text{rank}(T^2) + \text{nullity}(T^2).$$

Given $\text{rank}(T) = \text{rank}(T^2)$, it follows that $\text{nullity}(T) = \text{nullity}(T^2)$, which means $\dim(\text{Ker}(T)) = \dim(\text{Ker}(T^2))$. Clearly $\text{Ker}(T) \subseteq \text{Ker}(T^2)$ (since $T(x) = 0 \implies T^2(x) = 0$). Since their dimensions are equal and V is finite-dimensional, we know that $\text{Ker}(T) = \text{Ker}(T^2)$.

Returning to our element x , since $x \in \text{Ker}(T^2)$, we must have $x \in \text{Ker}(T)$. This implies $T(x) = 0$. But $y = T(x)$, so $y = 0$. Thus $\text{Ran}(T) \cap \text{Ker}(T) = \{0\}$. Using the Dimension Theorem again, $\dim(V) = \dim(\text{Ran}(T)) + \dim(\text{Ker}(T))$. From the definition of direct sum, if the intersection of two subspaces is zero and the sum of their dimensions equals the dimension of the space, their sum is direct. Therefore, $V = \text{Ran}(T) \oplus \text{Ker}(T)$.

Recall: Projections

- A linear operator $T : V \rightarrow V$ is called a **projection** if $T^2 = T$.
- If $V = W_1 \oplus W_2$, the projection on W_1 along W_2 is the operator T such that for $v = w_1 + w_2$ (with $w_1 \in W_1, w_2 \in W_2$), $T(v) = w_1$.

3. (Section 2.3, Exercise 17) Let V be a vector space. Determine all linear transformations $T : V \rightarrow V$ such that $T = T^2$. *Hint:* Note that $x = T(x) + (x - T(x))$ for every x in V , and show that $V = \{y : T(y) = y\} \oplus \text{Ker}(T)$.

Solution: Step 1: Identify the subspace of fixed points.

Let $W_1 = \{y \in V : T(y) = y\}$. If $y \in W_1$, then $y = T(y) \in \text{Ran}(T)$, so $W_1 \subseteq \text{Ran}(T)$. Conversely, if $y \in \text{Ran}(T)$, then $y = T(x)$ for some x . Applying T , we get $T(y) = T(T(x)) = T^2(x)$. Since $T = T^2$, we have $T(y) = T(x) = y$. Thus $\text{Ran}(T) \subseteq W_1$. Therefore, $\text{Ran}(T) = \{y : T(y) = y\}$.

Step 2: Show Intersection is Zero and sum is V

Let $v \in \text{Ran}(T) \cap \text{Ker}(T)$. Since $v \in \text{Ran}(T)$, $T(v) = v$ (from Step 1). Since $v \in \text{Ker}(T)$, $T(v) = 0$. Thus $v = 0$. So $\text{Ran}(T) \cap \text{Ker}(T) = \{0\}$. Let $x \in V$. Using the hint, we write $x = T(x) + (x - T(x))$. Let $y = T(x)$. Then $y \in \text{Ran}(T)$. Let $z = x - T(x)$. Then $T(z) = T(x - T(x)) = T(x) - T^2(x)$. Since $T = T^2$, $T(z) = T(x) - T(x) = 0$. So $z \in \text{Ker}(T)$. Thus, every vector can be written as a sum of a vector in $\text{Ran}(T)$ and a vector in $\text{Ker}(T)$. Therefore, $V = \text{Ran}(T) \oplus \text{Ker}(T)$, and T is the projection on $\text{Ran}(T)$ along $\text{Ker}(T)$.

Section 2.4: Invertibility and Isomorphisms

Recall: Isomorphisms

- A linear transformation $T : V \rightarrow W$ is an **isomorphism** if it is one-to-one and onto.
- **Theorem 2.6:** Let V and W be vector spaces, and let $\{v_1, \dots, v_n\}$ be a basis for V . For any vectors $w_1, \dots, w_n$ in W , there exists a unique linear transformation $T : V \rightarrow W$ such that $T(v_i) = w_i$ for all $i = 1 \dots n$.

4. (Section 2.4, Exercise 15) Let V and W be finite-dimensional vector spaces, and let $T : V \rightarrow W$ be a linear transformation. Suppose that β is a basis for V . Prove that T is an isomorphism if and only if $T(\beta)$ is a basis for W .

Solution: Let $\beta = \{v_1, v_2, \dots, v_n\}$. Then $T(\beta) = \{T(v_1), T(v_2), \dots, T(v_n)\}$.

($\Rightarrow$) Suppose T is an isomorphism. Since T is an isomorphism, it is one-to-one and onto. *Linear Independence:* Consider the linear combination $\sum_{i=1}^n c_i T(v_i) = 0$. By linearity, $T(\sum_{i=1}^n c_i v_i) = 0$. Since T is one-to-one, $\ker(T) = \{0\}$, so $\sum_{i=1}^n c_i v_i = 0$. Since β is a basis, it is linearly independent, so $c_i = 0$ for all i . Thus, $T(\beta)$ is linearly independent. *Spanning:* Let $w \in W$. Since T is onto, there exists $v \in V$ such that $T(v) = w$. Since β spans V , $v = \sum_{i=1}^n a_i v_i$. Then $w = T(\sum_{i=1}^n a_i v_i) = \sum_{i=1}^n a_i T(v_i)$. Thus, $T(\beta)$ spans W . Therefore, $T(\beta)$ is a basis for W .

($\Leftarrow$) Suppose $T(\beta)$ is a basis for W . *Onto:* Since $T(\beta)$ is a basis for W , $\text{span}(T(\beta)) = W$. Since the range of T contains the span of $T(\beta)$, $\text{Ran}(T) = W$. Thus T is onto. *One-to-one:* Let $v \in \ker(T)$. We can write $v = \sum_{i=1}^n c_i v_i$. $T(v) = T(\sum_{i=1}^n c_i v_i) = \sum_{i=1}^n c_i T(v_i) = 0$. Since $T(\beta)$ is a basis, it is linearly independent. Therefore, $c_i = 0$ for all i . This implies $v = 0$. Thus $\ker(T) = \{0\}$, so T is one-to-one. Since T is one-to-one and onto, it is an isomorphism.

Recall: Theorem 2.20 (Coordinate Isomorphism)

Let V and W be finite-dimensional vector spaces of dimensions n and m , respectively, and let β and γ be ordered bases for V and W , respectively. Then the function $\Phi : \mathcal{L}(V, W) \rightarrow \mathcal{M}_{m \times n}(F)$ defined by $\Phi(T) = [T]_{\beta}^{\gamma}$ is an isomorphism.

5. (Section 2.4, Exercise 21) Let V and W be finite-dimensional vector spaces with ordered bases $\beta = \{v_1, v_2, \dots, v_n\}$ and $\gamma = \{w_1, w_2, \dots, w_m\}$, respectively. By Theorem 2.6, there exist linear transformations $T_{ij} : V \rightarrow W$ such that

$$T_{ij}(v_k) = \begin{cases} w_i & \text{if } k = j \\ 0 & \text{if } k \neq j. \end{cases}$$

- (a) Let M^{ij} be the $m \times n$ matrix with 1 in the i th row and j th column and 0 elsewhere. Prove that $[T_{ij}]_{\beta}^{\gamma} = M^{ij}$.
- (b) Prove that $\{T_{ij} : 1 \leq i \leq m, 1 \leq j \leq n\}$ is a basis for $\mathcal{L}(V, W)$.

Solution: (a) The matrix representation $[T_{ij}]_{\beta}^{\gamma}$ is the $m \times n$ matrix where the k -th column is the coordinate vector $[T_{ij}(v_k)]_{\gamma}$. Consider the k -th column of this matrix:

$$T_{ij}(v_k) = \begin{cases} w_i & \text{if } k = j \\ 0 & \text{if } k \neq j. \end{cases}$$

- If $k \neq j$, then $T_{ij}(v_k) = 0$. The coordinate vector $[0]_{\gamma}$ is the zero vector.

- If $k = j$, then $T_{ij}(v_j) = w_i$. The coordinate vector $[w_i]_\gamma$ is the standard basis vector $e_i \in F^m$ (which has a 1 at index i and 0 elsewhere).

Thus, the matrix $[T_{ij}]_\beta^\gamma$ contains the vector e_i in the j -th column and zero vectors in all other columns. This describes precisely the matrix M^{ij} , which has a 1 at entry (i, j) and 0 elsewhere.

(b): Let $\Psi : \mathcal{L}(V, W) \rightarrow \mathcal{M}_{m \times n}(F)$ be the coordinate mapping defined by $\Psi(T) = [T]_\beta^\gamma$. We know that Ψ is an isomorphism by Theorem 2.20. Consequently, its inverse $\Psi^{-1} : \mathcal{M}_{m \times n}(F) \rightarrow \mathcal{L}(V, W)$ is also an isomorphism. We also know that the set of matrices $\mathcal{E} = \{M^{ij} : 1 \leq i \leq m, 1 \leq j \leq n\}$ is the standard basis for the vector space $\mathcal{M}_{m \times n}(F)$. From part (a), we established that $[T_{ij}]_\beta^\gamma = M^{ij}$, which can be written in terms of the map as $\Psi(T_{ij}) = M^{ij}$. Applying the inverse map to both sides, we get:

$$T_{ij} = \Psi^{-1}(M^{ij}).$$

We now apply the fact established in **Exercise 15** (specifically applied to the inverse map): An isomorphism maps a basis of the domain to a basis of the codomain. Since Ψ^{-1} is an isomorphism and $\mathcal{E} = \{M^{ij}\}$ is a basis for the domain $\mathcal{M}_{m \times n}(F)$, the image set $\{\Psi^{-1}(M^{ij})\} = \{T_{ij}\}$ must be a basis for the codomain $\mathcal{L}(V, W)$.

Section 2.5: The Change of Coordinate Matrix

Recall: Theorem 2.23 (Change of Basis)

Let T be a linear operator on a finite-dimensional vector space V , and let β and β' be ordered bases for V . Then $[T]_{\beta'} = Q^{-1}[T]_\beta Q$, where Q is the change of coordinate matrix from β' to β .

6. (Section 2.5, Exercise 4) Let T be the linear operator on $\mathbb{R}^2$ defined by

$$T \begin{bmatrix} a \\ b \end{bmatrix} = \begin{bmatrix} 2a + b \\ a - 3b \end{bmatrix}$$

Let β be the standard ordered basis for $\mathbb{R}^2$, and let

$$\beta' = \left\{ \begin{bmatrix} 1 \\ 1 \end{bmatrix}, \begin{bmatrix} 1 \\ 2 \end{bmatrix} \right\}$$

Use Theorem 2.23 and the fact that

$$\begin{bmatrix} 1 & 1 \\ 1 & 2 \end{bmatrix}^{-1} = \begin{bmatrix} 2 & -1 \\ -1 & 1 \end{bmatrix}$$

to find $[T]_{\beta'}$.

Solution:

Observe that the matrix representation relative to the standard basis is:

$$[T]_\beta = \begin{bmatrix} 2 & 1 \\ 1 & -3 \end{bmatrix}$$

According to Theorem 2.23, we have

$$[T]_{\beta'} = Q^{-1}[T]_\beta Q. \implies [T]_{\beta'} = \begin{bmatrix} 2 & -1 \\ -1 & 1 \end{bmatrix} \begin{bmatrix} 2 & 1 \\ 1 & -3 \end{bmatrix} \begin{bmatrix} 1 & 1 \\ 1 & 2 \end{bmatrix} = \begin{bmatrix} 8 & 13 \\ -5 & -9 \end{bmatrix}$$

Section 2.6: Dual Spaces

Recall: Dual Spaces and Bases

- **Dual Space:** For a vector space V over a field F , the dual space V^* is the vector space consisting of all linear functionals on V (linear transformations from V to F).
- **Dimension:** If V is finite-dimensional, then $\dim(V^*) = \dim(V)$.
- **Dual Basis:** If $\beta = \{v_1, v_2, \dots, v_n\}$ is an ordered basis for V , there exists a unique ordered basis $\beta^* = \{f_1, f_2, \dots, f_n\}$ for V^* called the **dual basis**, which satisfies the condition:

$$f_i(v_j) = \delta_{ij} = \begin{cases} 1 & \text{if } i = j \\ 0 & \text{if } i \neq j \end{cases}$$

- **Finding a Basis from a Dual Basis:** Conversely, if we are given a basis of functionals $\beta^* = \{f_1, \dots, f_n\}$ and need to find the basis $\beta = \{v_1, \dots, v_n\}$ of V for which it is the dual, we must find vectors v_j that satisfy the Kronecker delta condition $f_i(v_j) = \delta_{ij}$ for all i, j .

7. (Section 2.6, Exercise 4) Let $V = \mathbb{R}^3$, and define $f_1, f_2, f_3 \in V^*$ as follows:

$$f_1(x, y, z) = x - 2y, \quad f_2(x, y, z) = x + y + z, \quad f_3(x, y, z) = y - 3z.$$

Prove that $\{f_1, f_2, f_3\}$ is a basis for V^* , and then find a basis for V for which it is the dual basis.

Solution: Part 1: Prove that $\{f_1, f_2, f_3\}$ is a basis for V^* .

Since $\dim(V) = 3$, we know that $\dim(V^*) = 3$. To show that a set of 3 vectors is a basis for a 3-dimensional space, it suffices to show that the set is linearly independent.

Suppose we have a linear combination equal to the zero functional:

$$c_1 f_1 + c_2 f_2 + c_3 f_3 = \mathbf{0}.$$

This means that for any vector $(x, y, z) \in \mathbb{R}^3$:

$$c_1(x - 2y) + c_2(x + y + z) + c_3(y - 3z) = 0.$$

Evaluating this at the standard basis vectors of $\mathbb{R}^3$ we find $c_i = 0$. Hence the set $\{f_1, f_2, f_3\}$ is linearly independent and therefore forms a basis for V^* .

Part 2: Find the basis $\beta = \{v_1, v_2, v_3\}$ for V .

Solving three systems of linear equations $f_i(v_j) = \delta_{ij}$ for vectors v_1, v_2, v_3 we get the basis for V as

$$\beta = \left\{ \begin{pmatrix} 2/5 \\ -3/10 \\ -1/10 \end{pmatrix}, \begin{pmatrix} 3/5 \\ 3/10 \\ 1/10 \end{pmatrix}, \begin{pmatrix} 1/5 \\ 1/10 \\ -3/10 \end{pmatrix} \right\}.$$

Recall: Annihilators

- **Annihilator S^0 :** For a subset S of a vector space V , the annihilator S^0 is the set of all linear functionals $f \in V^*$ such that $f(x) = 0$ for all $x \in S$.
- **Double Dual Isomorphism ψ :** There is a canonical isomorphism $\psi : V \rightarrow V^{**}$ defined by $\psi(x)(f) = f(x)$ for all $f \in V^*$.

8. (Section 2.6, Exercise 14) Prove that if W is a subspace of V , then $\dim(W) + \dim(W^0) = \dim(V)$. *Hint:* Extend an ordered basis $\{x_1, x_2, \dots, x_k\}$ of W to an ordered basis $\beta = \{x_1, x_2, \dots, x_n\}$ of V . Let $\beta^* = \{f_1, f_2, \dots, f_n\}$. Prove that $\{f_{k+1}, f_{k+2}, \dots, f_n\}$ is a basis for W^0 .

Solution: Let $\beta = \{x_1, \dots, x_k, x_{k+1}, \dots, x_n\}$ where $\{x_1, \dots, x_k\}$ is a basis for W . Let $\beta^* = \{f_1, \dots, f_n\}$ be the dual basis, so $f_i(x_j) = \delta_{ij}$. Consider the set $B' = \{f_{k+1}, \dots, f_n\}$.

1. $B' \subseteq W^0$: For any f_j where $j > k$, and any $x \in W$, we can write $x = \sum_{i=1}^k c_i x_i$. Then $f_j(x) = \sum_{i=1}^k c_i f_j(x_i)$. Since $j > k$ and $i \leq k$, we have $i \neq j$, so $f_j(x_i) = \delta_{ji} = 0$. Thus $f_j(x) = 0$ for all $x \in W$, so $f_j \in W^0$.

2. Linear Independence: B' is a subset of a basis β^* , so it is linearly independent.

3. Spanning: Let $f \in W^0$. Since β^* is a basis for V^* , we can write $f = \sum_{i=1}^n a_i f_i$. For any $j \leq k$, consider $f(x_j)$. Since $x_j \in W$ and $f \in W^0$, we must have $f(x_j) = 0$. Using the expansion, $f(x_j) = \sum_{i=1}^n a_i f_i(x_j) = a_j$. So $a_j = 0$ for all $j \leq k$. Thus $f = \sum_{i=k+1}^n a_i f_i$, which means $f \in \text{span}(B')$.

Therefore B' is a basis for W^0 . The size of B' is $n - k$. Thus $\dim(W^0) = n - k$. Since $\dim(W) = k$ and $\dim(V) = n$, we have:

$$\dim(W) + \dim(W^0) = k + (n - k) = n = \dim(V).$$

9. (Section 2.6, Exercise 13c) Let V be a finite-dimensional vector space over F . For every subset S of V , define the annihilator S^0 of S as

$$S^0 = \{f \in V^* : f(x) = 0 \text{ for all } x \in S\}.$$

Prove that $(S^0)^0 = \text{span}(\psi(S))$, where ψ is defined as in Theorem 2.26.

Solution:

Recall from **Theorem 2.26** that $\psi : V \rightarrow V^{**}$ is the isomorphism defined by $\psi(x)(f) = f(x)$. We show that $\text{span}(\psi(S)) \subseteq (S^0)^0$. For any $f \in S^0$, and $x \in S$ we know $f(x) = 0$ by definition of the annihilator. Thus, $\psi(x)(f) = 0$. Since $\psi(x)$ maps every $f \in S^0$ to 0, $\psi(x) \in (S^0)^0$. Since $(S^0)^0$ is a subspace, it must contain the span of all such $\psi(x)$. Therefore, $\text{span}(\psi(S)) \subseteq (S^0)^0$.

Step 2: Dimension. We use the dimension formula for annihilators (Exercise 14): $\dim(W) + \dim(W^0) = \dim(V)$. Let $W = \text{span}(S)$. Then $S^0 = W^0$. Applying the formula to V : $\dim(S^0) = \dim(V) - \dim(\text{span}(S))$. Applying the formula to V^* : $\dim((S^0)^0) = \dim(V^*) - \dim(S^0)$. Since $\dim(V) = \dim(V^*)$, we substitute:

$$\dim((S^0)^0) = \dim(V) - [\dim(V) - \dim(\text{span}(S))] = \dim(\text{span}(S)).$$

Since ψ is an isomorphism, $\dim(\text{span}(\psi(S))) = \dim(\text{span}(S))$. Thus, the subspaces have equal dimension and one is contained in the other, so $(S^0)^0 = \text{span}(\psi(S))$.

1) Section 3.1. Question 2. Let

$$A = \begin{bmatrix} 1 & 2 & 3 \\ 1 & 0 & 1 \\ 1 & -1 & 1 \end{bmatrix}, \quad B = \begin{bmatrix} 1 & 0 & 3 \\ 1 & -2 & 1 \\ 1 & -3 & 1 \end{bmatrix}, \quad C = \begin{bmatrix} 1 & 0 & 3 \\ 0 & -2 & -2 \\ 1 & -3 & 1 \end{bmatrix}.$$

Find an elementary operation which transform A into B and an elementary operation which transform B into C. Transform C into I_3 .

Solution: We observe that

$$A \xrightarrow{-2C_1 + C_2} B \xrightarrow{-R_1 + R_2} C$$

Now let us transform C into I_3 .

$$C \xrightarrow{-\frac{1}{2} \cdot R_2} \begin{bmatrix} 1 & 0 & 3 \\ 0 & 1 & 1 \\ 1 & -3 & 1 \end{bmatrix} \xrightarrow{-R_1 + R_3} \begin{bmatrix} 1 & 0 & 3 \\ 0 & 1 & 1 \\ 0 & -3 & -2 \end{bmatrix} \xrightarrow{3R_2 + R_3} \begin{bmatrix} 1 & 0 & 3 \\ 0 & 1 & 1 \\ 0 & 0 & 1 \end{bmatrix}$$

$$\xrightarrow{-3R_3 + R_1} \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 1 \\ 0 & 0 & 1 \end{bmatrix} \xrightarrow{-R_3 + R_2} \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} = I_3.$$

② Section 3.1, Question 8:

Prove that if a matrix Q can be obtained from a matrix P by an elementary row operation, then P can be obtained from Q by an elementary row operation of the same type.

Solution:

Recall Theorem 3.2: Elementary matrices are invertible & the inverse of an elementary matrix is an elementary matrix of the same type.

So if Q can be obtained from P , we can write $Q = E \cdot P$

But then $E^{-1} \cdot Q = P$. Hence P can be obtained from Q by an elementary row operation of the same type.

③ Section 3.2. Question 5d

Compute the rank and the inverse of the following matrix if it exists.

$$A = \begin{bmatrix} 0 & -2 & 4 \\ 1 & 1 & -1 \\ 2 & 4 & -5 \end{bmatrix}$$

Solution:

We know from Theorem 3.5 that the rank of a matrix is the dimension of the subspace generated by its columns (or rows). And an $n \times n$ matrix is invertible if and only if its rank is n .

We perform Gaussian elimination on the augmented matrix $[A | I_3]$.

$$\left[\begin{array}{ccc|ccc} 0 & -2 & 4 & 1 & 0 & 0 \\ 1 & 1 & -1 & 0 & 1 & 0 \\ 2 & 4 & -5 & 0 & 0 & 1 \end{array} \right] \xrightarrow{R_1 \leftrightarrow R_2} \left[\begin{array}{ccc|ccc} 1 & 1 & -1 & 0 & 1 & 0 \\ 0 & -2 & 4 & 1 & 0 & 0 \\ 2 & 4 & -5 & 0 & 0 & 1 \end{array} \right]$$

$$\xrightarrow{-2R_1 + R_3 \rightarrow R_3} \left[\begin{array}{ccc|ccc} 1 & 1 & -1 & 0 & 1 & 0 \\ 0 & -2 & 4 & 1 & 0 & 0 \\ 0 & 2 & -3 & 0 & -2 & 1 \end{array} \right] \xrightarrow{R_2 + R_3 \rightarrow R_3} \left[\begin{array}{ccc|ccc} 1 & 1 & -1 & 0 & 1 & 0 \\ 0 & -2 & 4 & 1 & 0 & 0 \\ 0 & 0 & 1 & 1 & -2 & 1 \end{array} \right]$$

$\bar{A}$

$\bar{A}$ is an upper triangular matrix with non-zero diagonal entries.

$$\Rightarrow \text{rank}(A) = 3.$$

Thus there are 3 linearly independent rows.

$\Downarrow$
A is invertible.

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Continuing to row reduced echelon form we have

$$\left[\begin{array}{ccc|ccc} 1 & 1 & -1 & 0 & 1 & 0 \\ 0 & -2 & 4 & 1 & 0 & 0 \\ 0 & 0 & 1 & 1 & -2 & 1 \end{array} \right] \xrightarrow{-\frac{1}{2} \cdot R_2 \rightarrow R_2} \left[\begin{array}{ccc|ccc} 1 & 1 & -1 & 0 & 1 & 0 \\ 0 & 1 & -2 & -1/2 & 0 & 0 \\ 0 & 0 & 1 & 1 & -2 & 1 \end{array} \right]$$

$$\begin{array}{l} 2R_3 + R_2 \rightarrow R_2 \\ \xrightarrow{\hspace{1cm}} \\ R_3 + R_1 \rightarrow R_1 \end{array} \left[\begin{array}{ccc|ccc} 1 & 1 & 0 & 1 & -1 & 1 \\ 0 & 1 & 0 & 3/2 & -4 & 2 \\ 0 & 0 & 1 & 1 & -2 & 1 \end{array} \right]$$

$$\xrightarrow{-R_2 + R_1 \rightarrow R_1} \left[\begin{array}{ccc|ccc} 1 & 0 & 0 & -1/2 & 3 & -1 \\ 0 & 1 & 0 & 3/2 & -4 & 2 \\ 0 & 0 & 1 & 1 & -2 & 1 \end{array} \right]$$

$$\Rightarrow A^{-1} = \begin{bmatrix} -1/2 & 3 & -1 \\ 3/2 & -4 & 2 \\ 1 & -2 & 1 \end{bmatrix}$$

④ Section 3.2 Question 6a:

For the following linear transformation T , determine whether T is invertible & compute T^{-1} if it exists.

$$T: P_2(\mathbb{R}) \rightarrow P_2(\mathbb{R}) \text{ defined by } T(f(x)) = f''(x) + 2f'(x) + f(x)$$

Solution: Recall from Theorem 2.18's corollary, A linear transformation T is invertible if and only if its matrix representation $[T]_{\beta}$ is invertible.

Moreover from Theorem 2.14 we know $[T^{-1}]_{\beta} = ([T]_{\beta})^{-1}$.

Now let $\beta = \{1, x, x^2\}$ be the ordered basis for $P_2(\mathbb{R})$. Then

$$T(1) = -1$$

$$T(x) = 2 - x$$

$$T(x^2) = 2 + 4x - x^2$$

Therefore $[T]_{\beta} = \begin{bmatrix} -1 & 2 & 2 \\ 0 & -1 & 4 \\ 0 & 0 & -1 \end{bmatrix}$. Now we perform elementary row operations:

$$\left[[T]_{\beta} \mid I \right] \xrightarrow{\substack{-R_1 \\ -R_2 \\ -R_3}} \begin{bmatrix} 1 & -2 & -2 \\ 0 & 1 & -4 \\ 0 & 0 & 1 \end{bmatrix} \xrightarrow{\substack{4R_3 + R_2 \rightarrow R_2 \\ 2R_3 + R_1 \rightarrow R_1}} \begin{bmatrix} 1 & -2 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} \xrightarrow{2R_2 + R_1 \rightarrow R_1} I_3.$$

Therefore applying the same operations to I_3 we get $[T^{-1}]_{\beta} = \begin{bmatrix} -1 & -2 & -10 \\ 0 & -1 & -4 \\ 0 & 0 & -1 \end{bmatrix}$

$$\text{Hence } T^{-1}(a+bx+cx^2) = (-a-2b-10c) + (-b-4c)x + (-c)x^2.$$

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⑤ Section 3.2. 14:

Let $T, U: V \rightarrow W$ be linear transformations.

(a) Prove that $\text{Ran}(T+U) \subseteq \text{Ran}(T) + \text{Ran}(U)$

(b) Prove that if W is finite dimensional, then $\text{rank}(T+U) \leq \text{rank}(T) + \text{rank}(U)$.

(c) Deduce from (b) that $\text{rank}(A+B) \leq \text{rank}(A) + \text{rank}(B)$ for any $m \times n$ matrices A and B .

Solution:

(a) Pick $y \in \text{Ran}(T+U)$. Then there exists $x \in V$: $(T+U)(x) = y$.

By definition of addition of linear transformations $(T+U)(x) = T(x) + U(x)$.

Since $T(x) \in \text{Ran}(T)$, and $U(x) \in \text{Ran}(U)$, it follows that $y \in \text{Ran}(T) + \text{Ran}(U)$.

Hence $\text{Ran}(T+U) \subseteq \text{Ran}(T) + \text{Ran}(U)$.

(b) By part (a), $\text{Ran}(T+U)$ is a subspace of $\text{Ran}(T) + \text{Ran}(U)$. Thus

$$\dim(\text{Ran}(T+U)) \leq \dim(\text{Ran}(T)) + \dim(\text{Ran}(U))$$

Actually we have

$$\dim(\text{Ran}(T+U)) \leq \dim(\text{Ran}(T) + \text{Ran}(U)) = \dim(\text{Ran}(T)) + \dim(\text{Ran}(U)) - \dim(\text{Ran}(T) \cap \text{Ran}(U)) \leq \dim(\text{Ran}(T)) + \dim(\text{Ran}(U))$$

$$\text{rank}(T+U) \leq \text{rank}(T) + \text{rank}(U).$$

(c) Let $A, B \in M_{m \times n}(F)$. Let $L_A, L_B : F^n \rightarrow F^m$ be the left multiplication transformations. We know by definition that

$$\text{rank}(A) = \text{rank}(L_A), \text{ and } \text{rank}(B) = \text{rank}(L_B).$$

Moreover $L_{A+B} = L_A + L_B$. Applying part (b) to L_A and L_B

we get

$$\text{rank}(L_{A+B}) \leq \text{rank}(L_A) + \text{rank}(L_B)$$

$$\Rightarrow \text{rank}(A+B) \leq \text{rank}(A) + \text{rank}(B).$$

6) Section 3.3 Question 4 (a): Consider the system

$$x_1 + 3x_2 = 4$$

$$2x_1 + 5x_2 = 3.$$

* Compute A^{-1} where A is the invertible coefficient matrix of the system.

* Use A^{-1} and solve the system.

Solution: Recall from Theorem 3.10 that if A is invertible then the system $A \cdot x = b$, has exactly one solution, namely $x = A^{-1} \cdot b$.

The given system is $A \cdot x = b$ with $A = \begin{bmatrix} 1 & 3 \\ 2 & 5 \end{bmatrix}$, and $b = \begin{bmatrix} 4 \\ 3 \end{bmatrix}$.

We first compute A^{-1} by forming the augmented matrix and perform the row reduction.

$$\left[\begin{array}{cc|cc} 1 & 3 & 1 & 0 \\ 2 & 5 & 0 & 1 \end{array} \right] \xrightarrow{-2R_1 + R_2 \rightarrow R_2} \left[\begin{array}{cc|cc} 1 & 3 & 1 & 0 \\ 0 & -1 & -2 & 1 \end{array} \right]$$

$$\xrightarrow{-R_2 \rightarrow R_2} \left[\begin{array}{cc|cc} 1 & 3 & 1 & 0 \\ 0 & 1 & 2 & -1 \end{array} \right] \xrightarrow{-3R_2 + R_1 \rightarrow R_1} \left[\begin{array}{cc|cc} 1 & 0 & -5 & 3 \\ 0 & 1 & 2 & -1 \end{array} \right].$$

$$\text{Thus } A^{-1} = \begin{bmatrix} -5 & 3 \\ 2 & -1 \end{bmatrix}.$$

Now the solution to the system is

$$x = A^{-1} \cdot b = \begin{bmatrix} -5 & 3 \\ 2 & -1 \end{bmatrix} \cdot \begin{bmatrix} 4 \\ 3 \end{bmatrix} = \begin{bmatrix} -11 \\ 5 \end{bmatrix}. \quad \text{Hence } x_1 = -11, \quad x_2 = 5$$

(9)

7 Section 3.3, Question 7(c). Decide if the following system has a solution:

$$x_1 + x_2 - x_3 = 0.$$

$$x_1 + 2x_2 + x_3 = 3$$

$$x_1 + 2x_2 - x_3 = 1.$$

Solution: Recall from Theorem 3.11 that the system $A \cdot x = b$ has a solution if and only if $\text{rank}(A) = \text{rank}(A|b)$.

We construct the augmented matrix $[A|b]$ and reduce it to determine the rank.

$$\left[\begin{array}{ccc|c} 1 & 1 & -1 & 0 \\ 1 & 2 & 1 & 3 \\ 1 & 2 & -1 & 1 \end{array} \right] \xrightarrow[\substack{-R_1+R_2 \rightarrow R_2 \\ -R_1+R_3 \rightarrow R_3}]{-R_1+R_2 \rightarrow R_2} \left[\begin{array}{ccc|c} 1 & 1 & -1 & 0 \\ 0 & 1 & 2 & 3 \\ 0 & 1 & 0 & 1 \end{array} \right] \xrightarrow{-R_2+R_3 \rightarrow R_3} \left[\begin{array}{ccc|c} 1 & 1 & -1 & 0 \\ 0 & 1 & 2 & 3 \\ 0 & 0 & -2 & -2 \end{array} \right]$$

Now we have the row echelon form & the rank of the coefficient matrix is 3. We also see the rank of the augmented matrix is 3. Since $\text{rank}(A) = \text{rank}(A|b)$. The system has a solution.

⑧ Section 3.3 Question 8:

Let $T: \mathbb{R}^3 \rightarrow \mathbb{R}^3$ be defined by

$$T(a, b, c) = (a+b, b-2c, a+2c).$$

For each vector $v \in \mathbb{R}^3$, determine if $v \in \text{Ran}(T)$.

(a) $v = (1, 3, -2)$, (b) $v = (2, 1, 1)$.

Solution: We may simply solve the system $\begin{matrix} a+b=1 \\ b-2c=3 \\ a+2c=-2 \end{matrix}$ if possible. (for part (a)).

Equivalently let A be the representing matrix of T . We have

$$\begin{aligned} T(e_1) &= (1, 0, 1) \\ T(e_2) &= (1, 1, 0) \\ T(e_3) &= (0, -2, 2) \end{aligned} \Rightarrow A = \begin{bmatrix} 1 & 1 & 0 \\ 0 & 1 & -2 \\ 1 & 0 & 2 \end{bmatrix}.$$

We know that $A \cdot x = v$ has a solution if and only if $v \in \text{Ran}(LA)$. We test the consistency for $A \cdot x = v$ by row reducing $[A|v]$.

$$\left[\begin{array}{ccc|c} 1 & 1 & 0 & v_1 \\ 0 & 1 & -2 & v_2 \\ 1 & 0 & 2 & v_3 \end{array} \right] \xrightarrow{-R_1 + R_3 \rightarrow R_3} \left[\begin{array}{ccc|c} 1 & 1 & 0 & v_1 \\ 0 & 1 & -2 & v_2 \\ 0 & -1 & 2 & v_3 - v_1 \end{array} \right]$$

$$\xrightarrow{R_2 + R_3 \rightarrow R_3} \left[\begin{array}{ccc|c} 1 & 1 & 0 & v_1 \\ 0 & 1 & -2 & v_2 \\ 0 & 0 & 0 & v_3 - v_1 + v_2 \end{array} \right]$$

For the system to be consistent we must have

$$v_3 - v_1 + v_2 = 0.$$

If $v = (1, 3, -2)$, then $-2 - 1 + 3 = 0 \Rightarrow v \in \text{Ran}(T)$.
If $v = (2, 1, 1)$, then $1 - 2 + 1 = 0 \Rightarrow v \in \text{Ran}(T)$.

g) T/F. If the coefficient matrix of a system of m linear equations in n unknowns has rank m , then the system has a solution.

Solution: Let the system be $A \cdot x = b$ where A is an $m \times n$ matrix. Consider

$$L_A: \mathbb{F}^n \rightarrow \mathbb{F}^m$$
$$x \mapsto A \cdot x$$

We are given $\text{rank}(A) = m = \dim(\text{Ran}(L_A))$.

We also have $\dim(\mathbb{F}^m) = m$. Hence $\text{Ran}(L_A) = \mathbb{F}^m$.

$\therefore L_A$ is onto. Therefore $\forall b \in \mathbb{F}^m \exists x \in \mathbb{F}^n : A \cdot x = b$.

So the system has a solution.